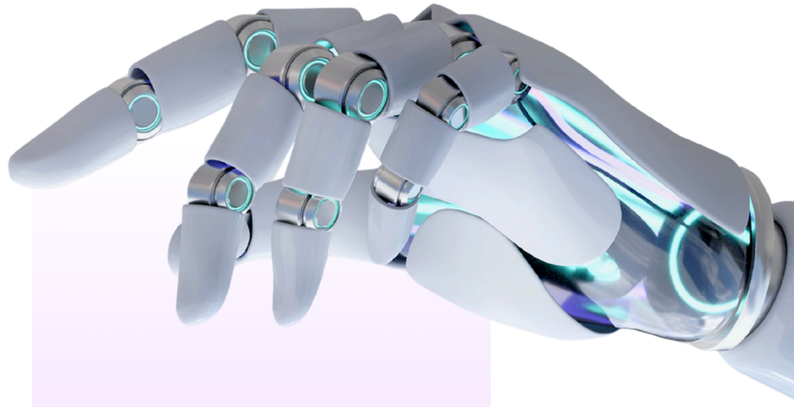
BOUIRA UNIVERSITY

Department of Computer Science.



HUMAN MACHINE INTERFACE



Presented by
Dr S.BENBATATA

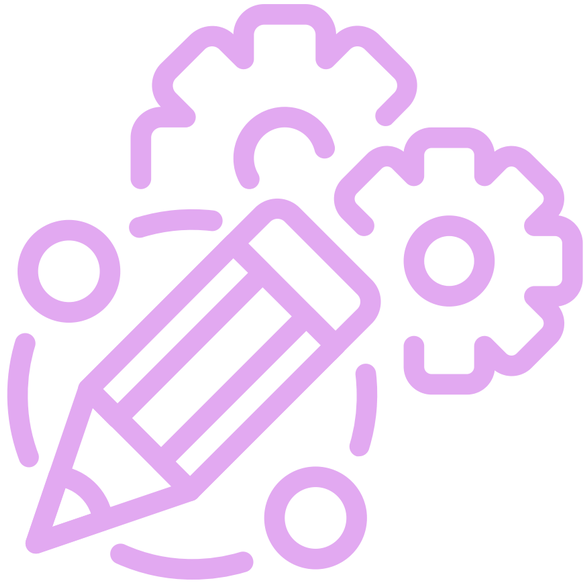
WHAT IS A HUMAN-MACHINE INTERFACE ?



Module Objectives



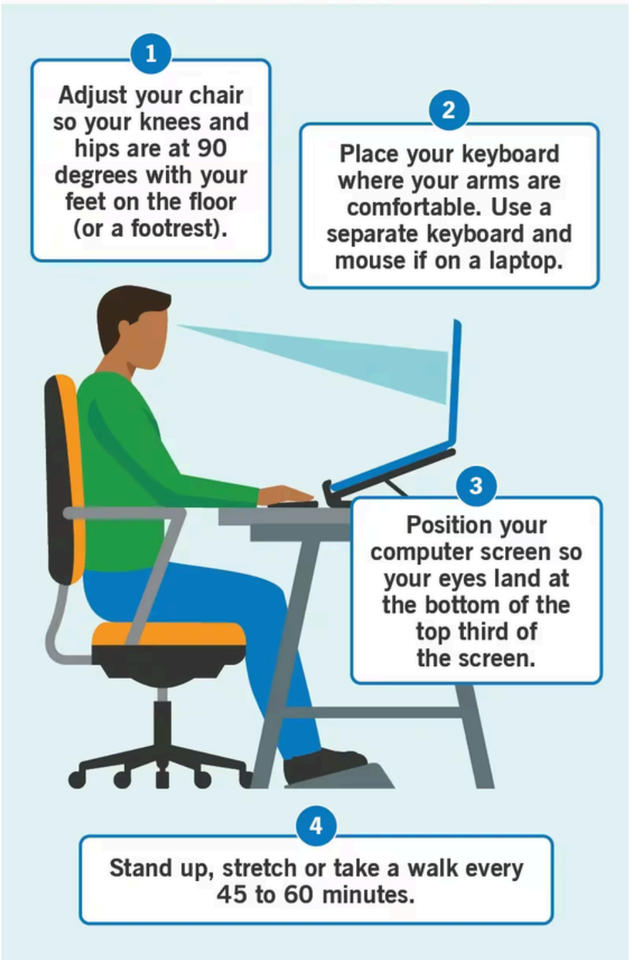
- Become familiar with interface design approaches.



Module Objectives

- Learn to design Human-Machine Interfaces (HMI) in accordance with ergonomic guidelines.





https://health.clevelandclinic.org/heres-how-to-set-up-your-office-to-avoid-aches-pain

Module Objectives

- To help future user interface designers develop the concepts and strategies needed to make good design decisions.

https://health.clevelandclinic.org/heres-how-to-set-up-your-office-to-avoid-aches-pain

Module Objectives

- To stress the importance of good user interface design.

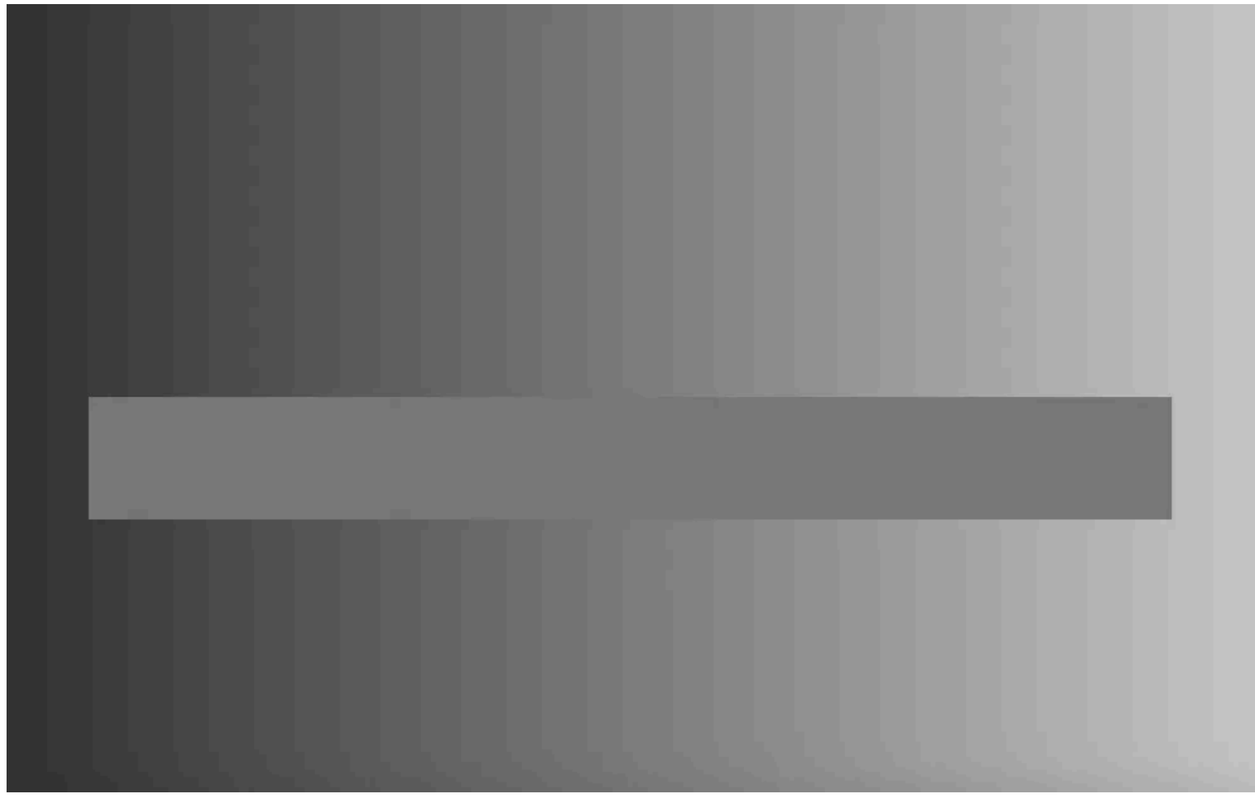
- To expose the future user interface designer to tools, techniques, and ideas for interface design.

Module Objectives

- To stress the importance of good user interface design.

Human-Computer Interaction. You may have a sense of some of the material since you are humans, technology users, and computer students!

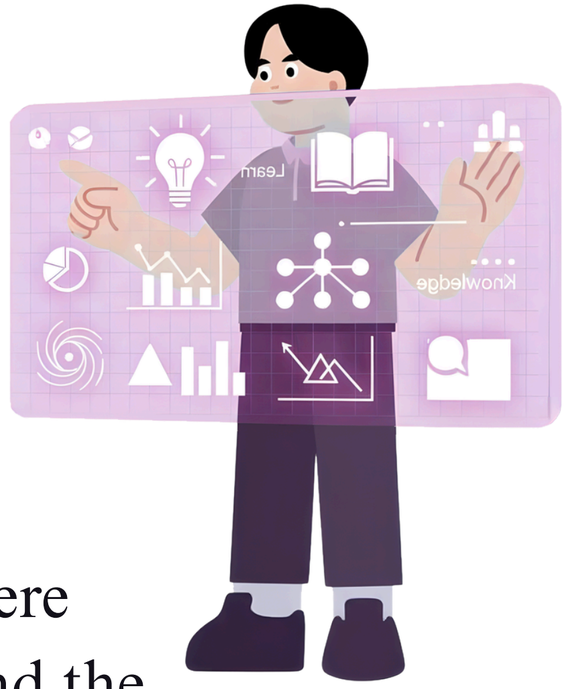
Example: Humans can distinguish millions of colors, but only few, about 30, shades of gray. Optical illusion. in the image?





Course 1: Introduction to HMI

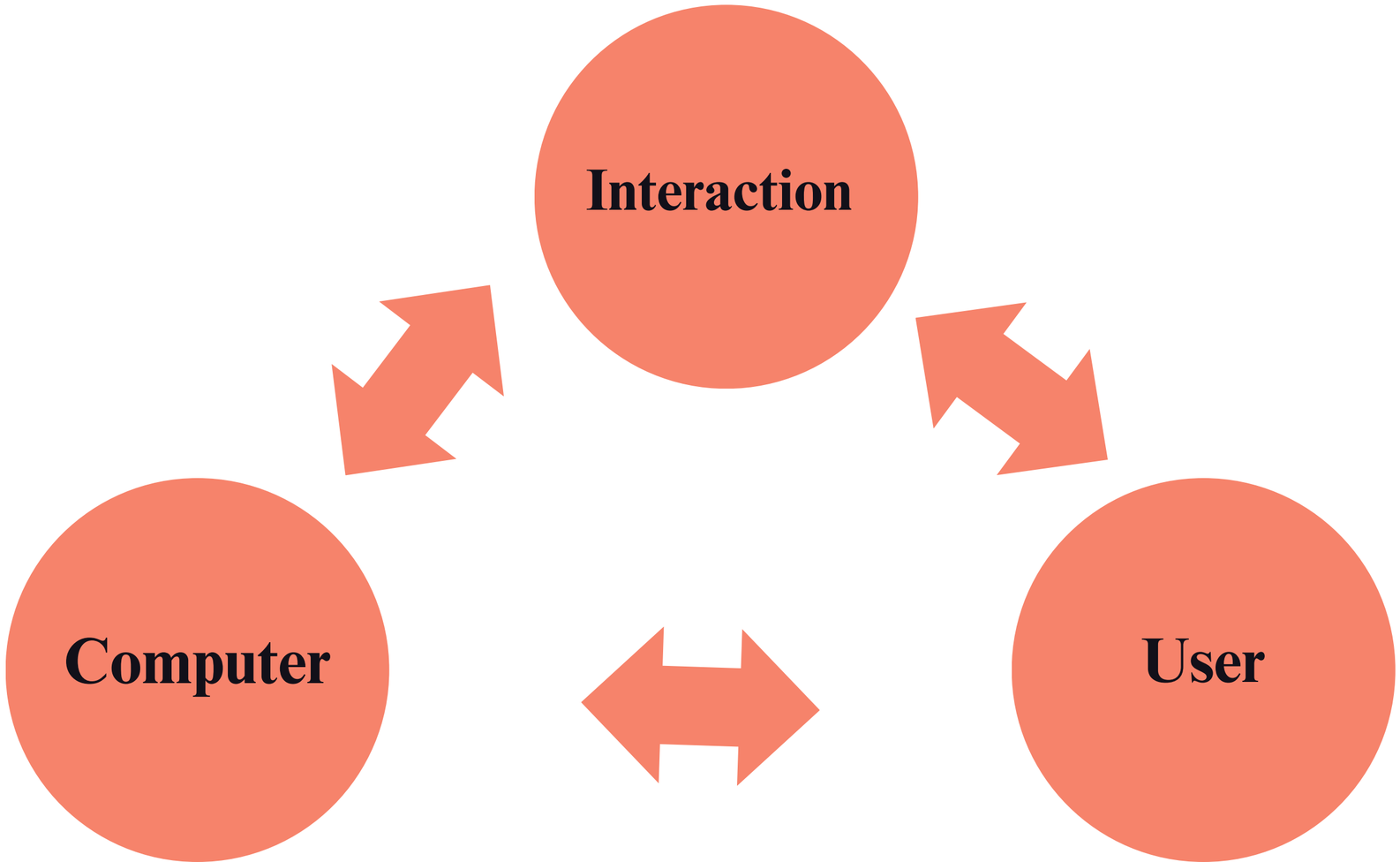
INTRODUCTION TO HCI



- Computer Human Interface: HCI is a field of study that refers to communication between the human user and a computer system. Here interface refers to a medium or interaction between the computer and the end user. It is also known as CHI (Human Computer Interaction) or MMI (Man Machine Interaction). It is concerned with design, evaluation, and implementation. It is used to provide a user-friendly environment

- User interface, design is a subset of a field of study called human-computer interaction (HCI).

INTRODUCTION TO HCI



INTRODUCTION TO HCI

User

By "user", we may mean an individual user, a group of users working together. Different users think about interaction in different ways. They also learn and remember things differently. Culture and country also make a difference.

Computer

When we talk about the computer, we're referring to any technology ranging from desktop computers, to large scale computer systems.
For example, if we were discussing the design of a Website, then the Website itself would be referred to as "the computer". Devices such as mobile phones can also be considered to be —computers.

INTRODUCTION TO HCI

Interaction

a two-way exchange between a human and a machine

The human does something → the machine responds → the human sees/hears the response → the human reacts again.

Example:

ATM interaction:

- You insert your card.
- The machine asks for your PIN.
- You enter the PIN.
- The machine gives you cash.

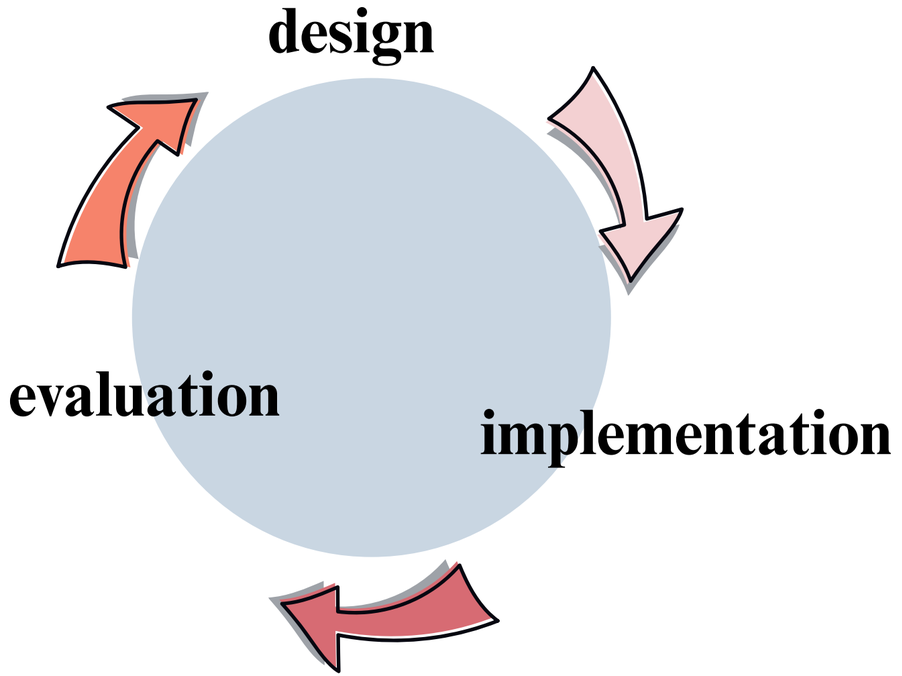
INTRODUCTION TO HCI

This field studies how humans interact with computers, or with one another through computers, and how to design systems that are ergonomic, efficient, easy to use, and generally adapted to their context of use.

CHI Definition

The following definition is given by the Association for Computing Machinery

> “Human-computer interaction is a discipline concerned with the design, evaluation, and implementation of interactive computing systems for human use, and with the study of the major phenomena related to them.”



The user interface is to the part of a computer and its software that people can see, hear, touch, talk to, etc.

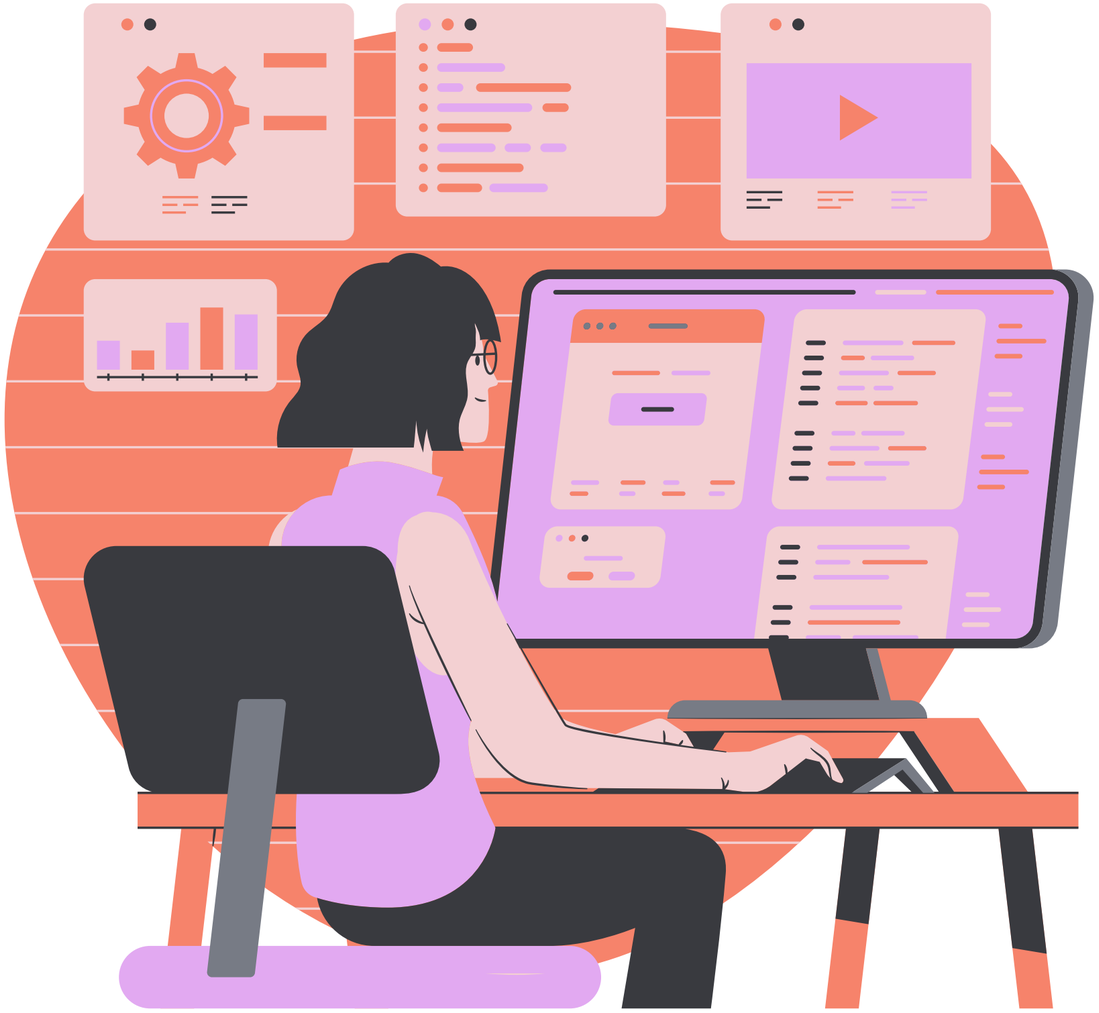
The Goals of HCI

The goals of HCI are to produce usable and safe systems, as well as functional systems.

In order o produce computer systems with good usability, developers must attempt to: understand the factors that determine how people use technology, develop tools and techniques to enable building suitable systems, achieve efficient, effective, and safe interaction put people first.

The Goals of HCI

Usability?



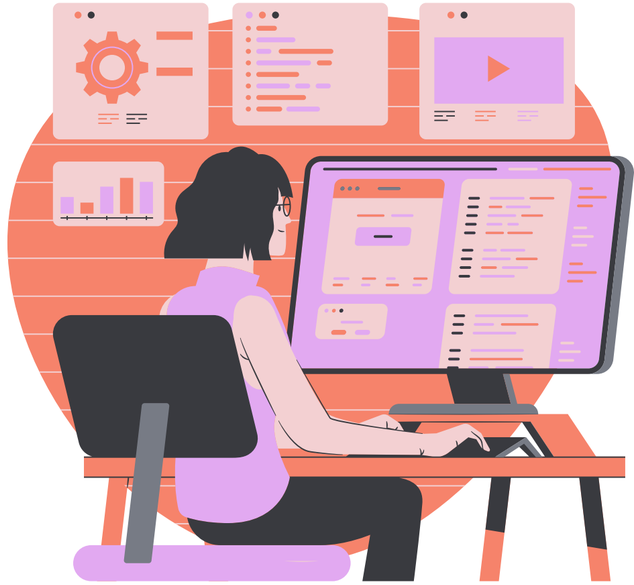
The Goals of HCI

Usability

Usability is one of the key concepts in IHM. It is concerned with making systems easy to learn and use.

A usable system is:

- easy to learn
- easy to remember how to use
- effective to use
- efficient to use
- safe to use
- enjoyable to use

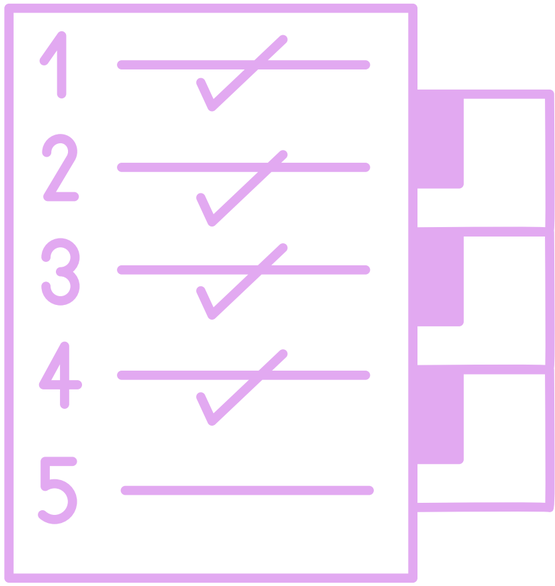


The Goals of HCI

efficient, effective

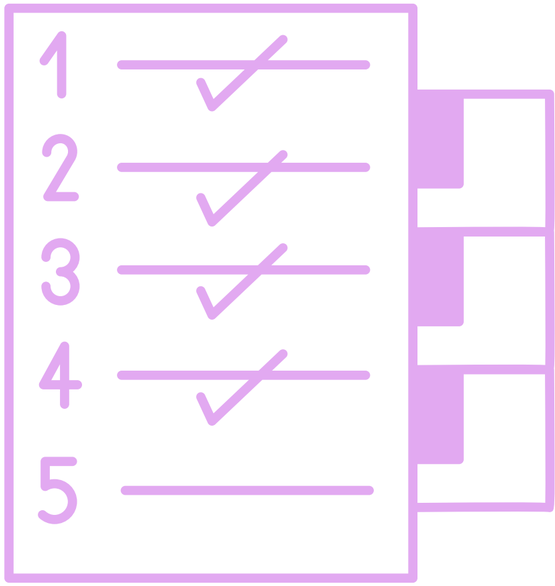


HCI designers must consider a variety of factors:



HCI designers must consider a variety of factors:

- What people want and expect, physical limitations and abilities people possess,
- How information processing systems work
- What people find enjoyable and attractive.
- Technical characteristics and limitations of the computer hardware and software must also be considered.



User interface

HMI is concerned not only with graphical interfaces but also with input/output (I/O) devices.

Input is how a person communicates his / her needs to the computer

- Some common input components are the keyboard, mouse, trackball, one's finger, and one's voice,QR code.

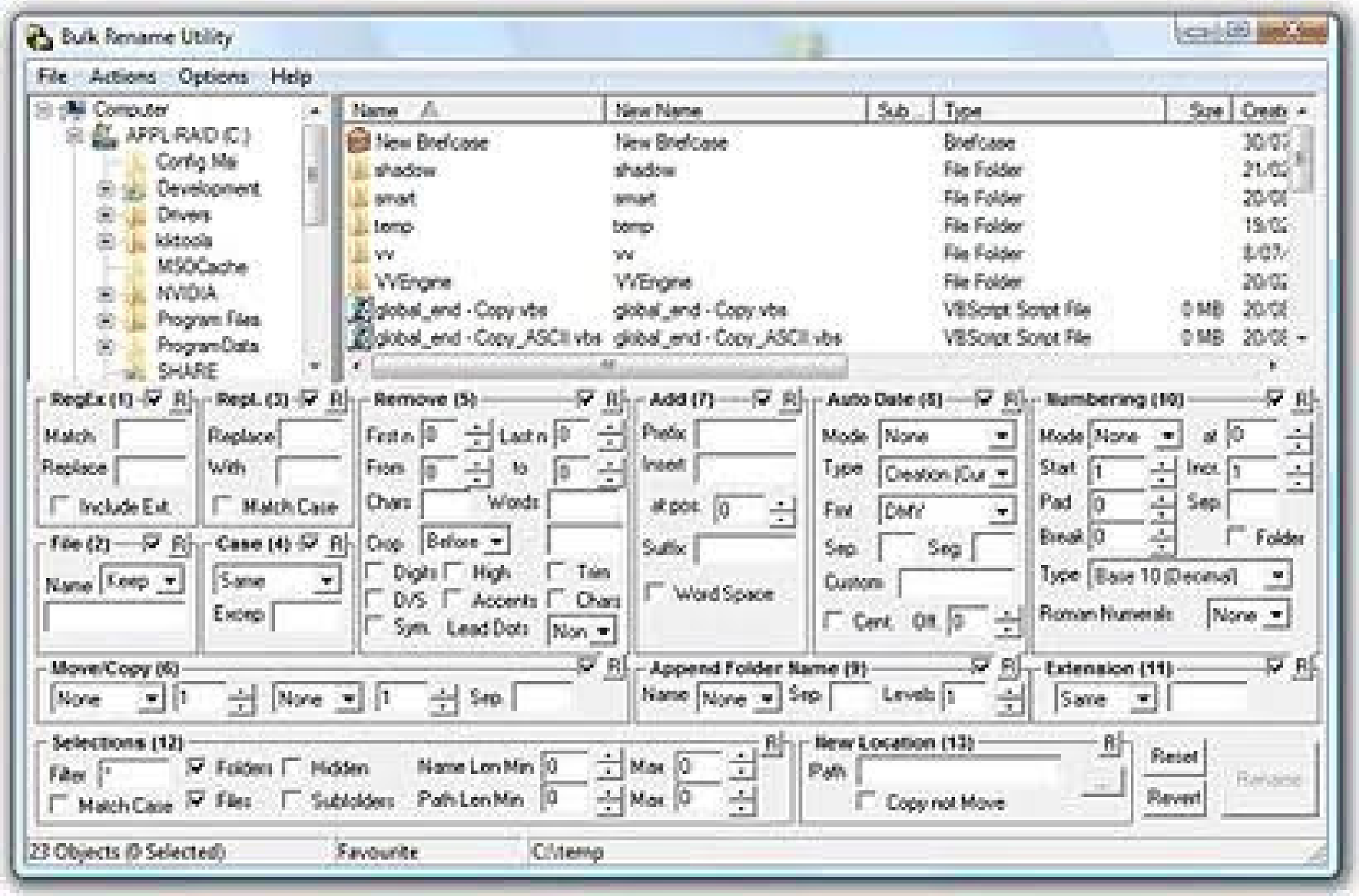
Output is how a computer gives results and information back to the user

- The screen.
- output through speakers, headphones, printers, projectors, vibration, lights, etc.,

Common mistakes by developers:

- They only care about how the system works.
- They design the interface at the end.
- They treat the interface as just a product feature.
- They think all users are like them.
- They never talk to real users.

The Risks of a Poor HMI



The Risks of a Poor HMI

- User errors: wrong actions, wrong data entry
- Safety risks :accidents in cars, factories, medical devices
- Slower work: more time and effort to complete tasks
- Frustration and stress :users dislike the system
- Low productivity
- System rejection : users refuse to use it
- Higher training and support costs : users need more help
- High maintenance cost.

HMI and Other Sciences

HCI covers many topics. It has grown with help from many other fields. The main fields that have contributed to HCI are:

HMI and Other Sciences

HCI covers many topics. It has grown with help from many other fields. The main fields that have contributed to HCI are:

Computer Science

- technology
- software design, development & maintenance
- User Interface Management Systems & User Interface Development Environments (UIDE)
- prototyping tools
- graphics

HMI and Other Sciences

Cognitive Psychology

- information processing
- capabilities
- limitations
- cooperative working
- performance prediction

HMI and Other Sciences

Social Psychology

- social & organizational structures

Ergonomics/Human Factors

- hardware design
- display readability

HMI and Other Sciences

Linguistics

- natural language interfaces

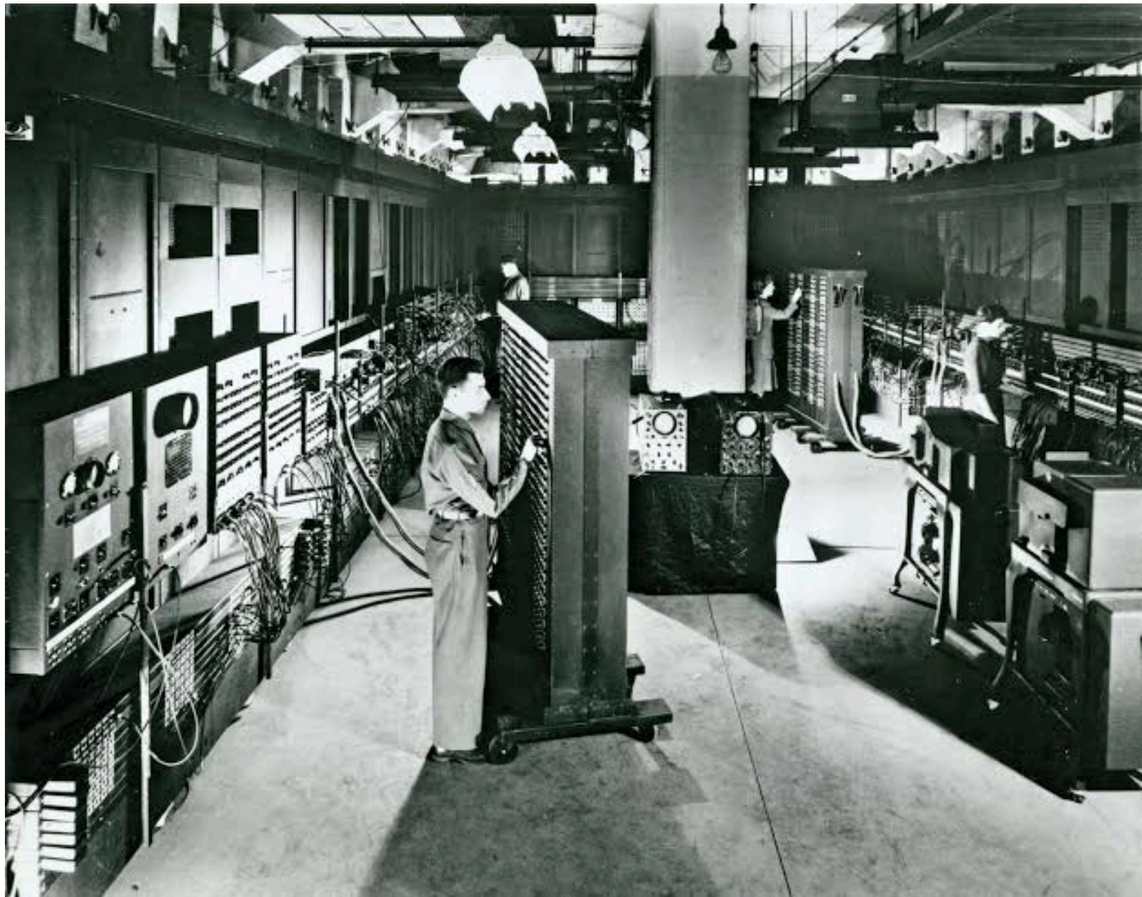
Artificial Intelligence

- intelligent software Engineering & Design
- graphic design

History of HMI

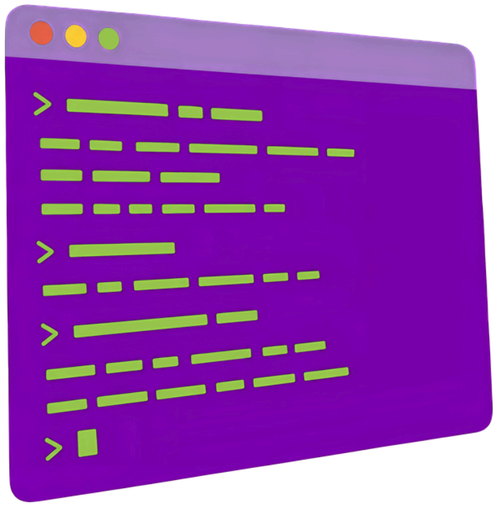
- Early computers was extremely difficult to use, it was large and expensive.

  It was used by specialists or engineers.



History of HMI

- In mid 1960's command line interface(CLI) was used to interact with computer. CLI are light weight and requires few memory consumption



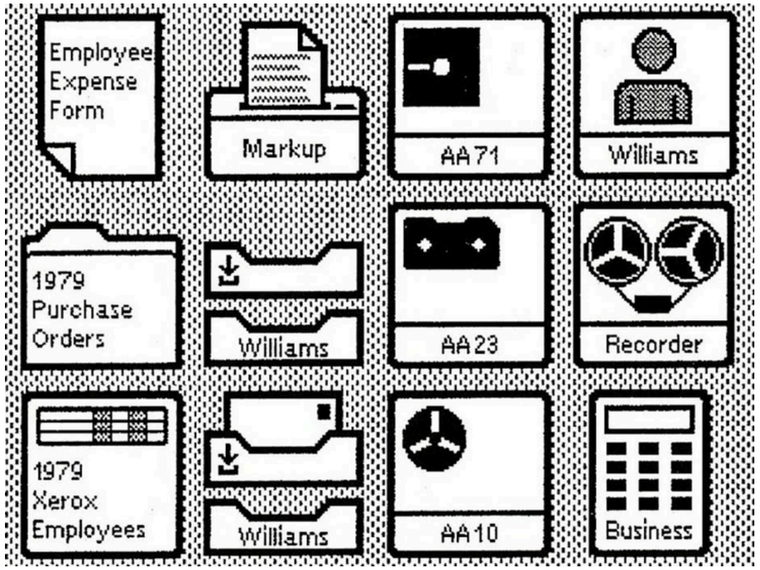
Apple II (1977) – one of the first produced personal computers



History of HMI

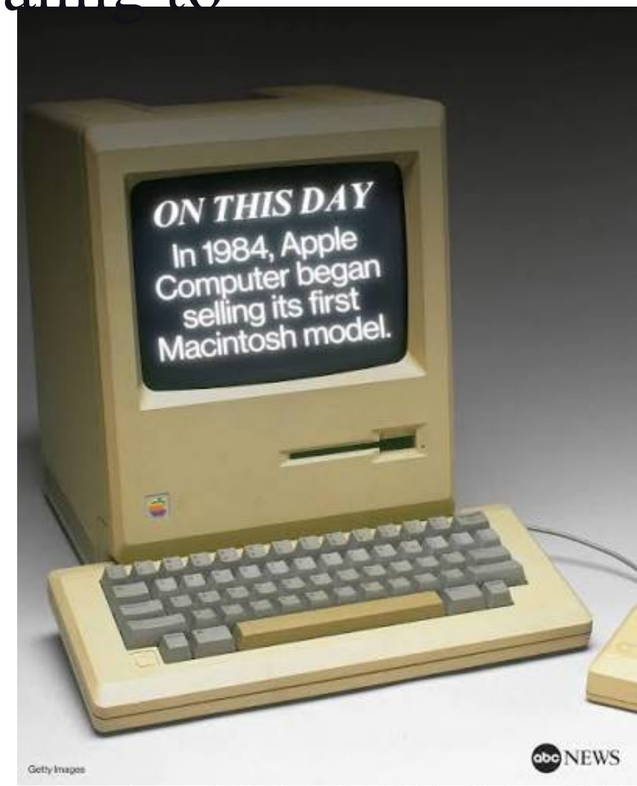
- 1980's are the booming phase for IHM. Some of the market leaders like Apple and Microsoft plays a crucial role for the modern development of HCI.
- GUI (Graphic User Interface) application was created that was easy to use, understand and visualize.
- Emergence of the WYSIWYG concept: use of windows and icon

WYSIWYG = What You See Is What You Get.



History of HMI

- In 1984 first Macintosh was release. It had good graphic user interface and a variety of fonts that makes your document more appealing to used.



Your paragraphhttps://www.facebook.com/ABCWorldNewsNow/posts/on-this-day-in-1984-apple-computer-began-selling-its-first-macintosh-modelit-boa/1026804686144729/ t

History of HMI

In 1990's internet starts. Communication between people become very easy through social networking like Email. The World Wide Web(WWW)is way for people to share information

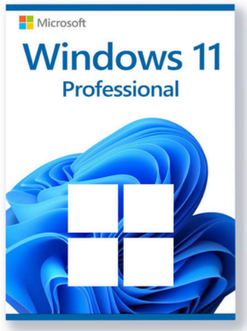
History of HMI

In 2000 mobile, laptop, tablet was a buzz word in this period. User can connect with anyone at any place. Smart phones comes into picture. User don't need any mouse or pointing device to select anything.

History of HMI

Micosoft Windows:





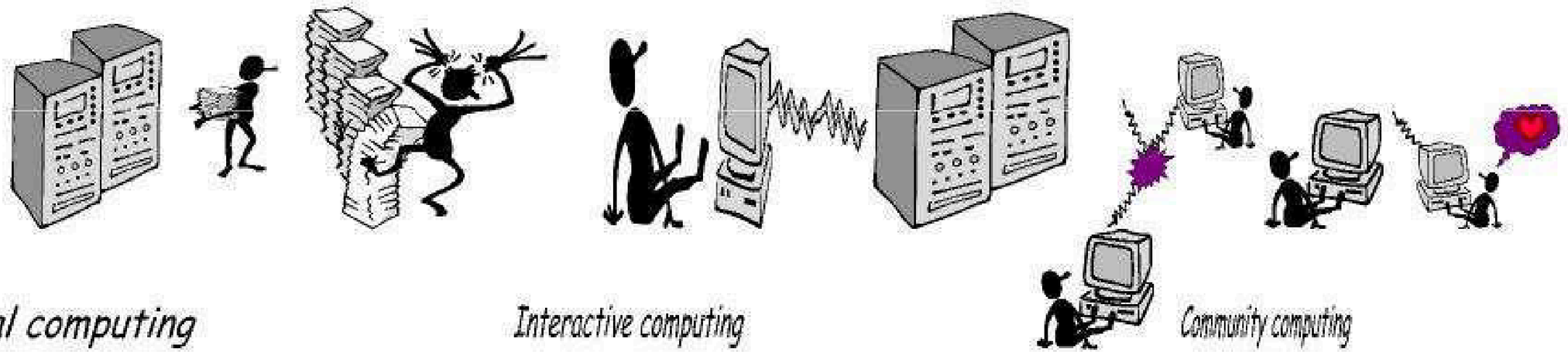


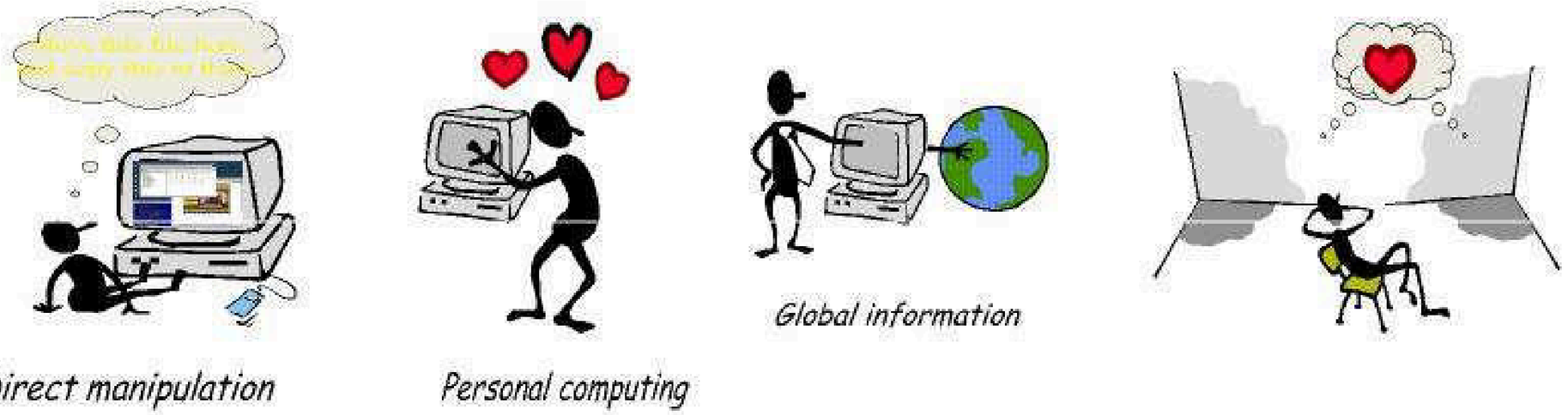
Development of PCs and laptops







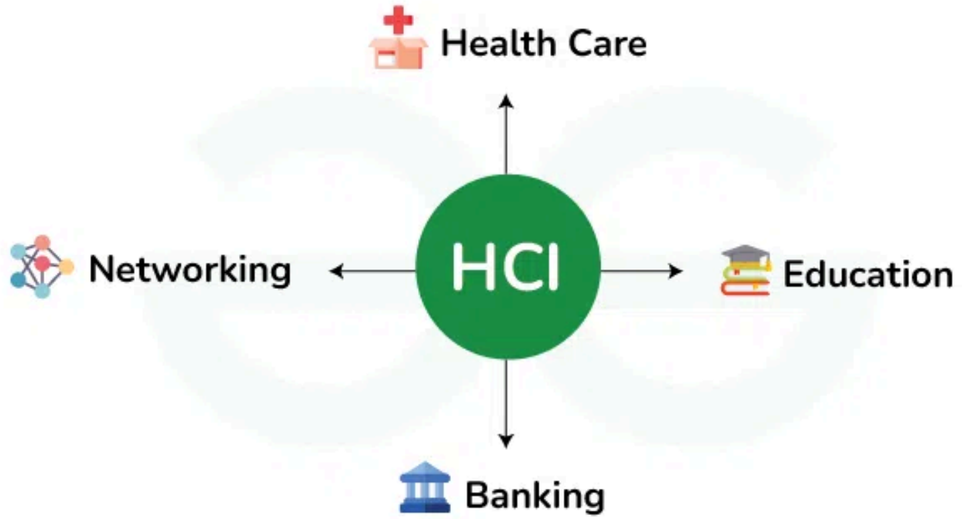




Human-Machine Interface (HMI) refers to the set of technical devices (hardware and software) that enable a human to control and communicate with a machine.

Application of HCI in different domains:

It includes the design and development of application. These application includes desktop application, websites and mobile apps. These application are used in different domains it includes healthcare, banking, education, networking and many more.



Applications of HCI

Application of HCI in different domains:

Health care

Patients have so many options now a days. They can buy medicines online and contact with doctor just with the help of mobile application.

Now doctor can use 3D animations to visualize the process.



Application of HCI in different domains:

Education

Now students can understand any concept more easily. There are so many resources available on internet. Class room teaching are now very interesting with the help of smart classes. Students have option to study online.

During COVID-19 students couldn't able to go outside their home. In this situation they have option to study online.



Application of HCI in different domains:

Banking

Now common people don't need to wait in a bank. They can get banking solution right at their home using Net banking or Mobile banking.

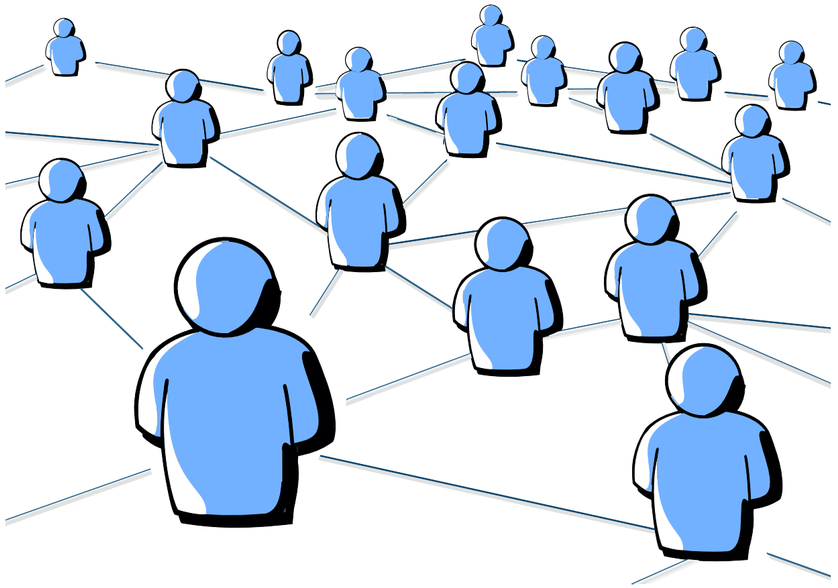


Application of HCI in different domains:

Networking

It includes social media networking and business networking.

Now it is very easy for us to connect and share information



CONCLUSION